

Bharath Pillai

614-687-6451 | bharathpotech@gmail.com | [linkedin.com/in/BharathPillai](https://www.linkedin.com/in/BharathPillai) | github.com/Bharathpillai06

EDUCATION

The Ohio State University

B.S. in Computer Science & Engineering | Minor: Robotics and Autonomous Systems

Columbus, OH

Expected May 2028

SKILLS

Programming Languages: Python, Java, Rust, C, Assembly

Frameworks: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, Matplotlib

Developer Tools: Git, Docker, AWS, Raspberry Pi, ESP 32, VS Code

Relevant Coursework: Data Structures & Algorithms, Database Systems, Electronics for CS, Linear Algebra

EXPERIENCE

National Science Foundation ICICLE AI Institute

Columbus, OH

Undergraduate Researcher – Computer Vision & Edge Systems

May 2025 – Present

- Built a model to get detected animals GPS coordinates using HFOV, camera bearing, and Vincenty geodesic math
- Validated GPS localization results against acoustic data, demonstrating cross-modal agreement with ~80% overlap
- Merged multi modal audio and camera data using Pandas, NumPy, and OpenCV reducing reporting time by ~70%

Ohio State University Dept. of Engineering & National Security Agency

Columbus, OH

Undergraduate Researcher – Software Defined Radio Systems

Mar 2026 – May 2026

- Implemented a SPECK128/256 cipher for an anti-jam frequency algorithm, tested against NSA vectors
- Developed diagrams for frequency assignment and bitsliced index computation for multi-user channel hopping
- Deployed the radio achieving 10,000 hops per second with costs less than \$5,000

OnRamp Hub: Ohio | Defense Innovation Unit (DIU) Program

Columbus, OH

Subject Matter Expert Intern

May 2026 – Present

- Evaluating early-stage technologies from vendors, creating technical capability summaries for DoW partners
- Meeting and researching emerging critical tech, maintaining a database of companies and R&D trends

PROJECTS

Wildlife Tracking Drone | Python, YOLOv8, TensorRT, DeepSORT, MAVLink, RealSense

Present

- Building an autonomous drone that tracks wildlife using an onboard Jetson Nano
- Optimized a YOLOv8 + TensorRT detection pipeline to 60 FPS with DeepSORT multi-target tracking
- Fused RealSense D435i depth and optical flow to estimate target range and velocity
- Implemented a PID controller driving a Pixhawk 6C over UART

BioHack 2nd Place (200 participants) | Python, PyTorch, OpenCV, Blender

Feb 2026

- Created a deep learning workflow for analysis across heterogeneous kidney imaging
- Preprocessed medical imagery with OpenCV by segmentation and masking for clear inputs for model inference
- Modeled kidney cortex using BPY in Blender, creating NURBS curve tubules with procedural sinusoidal winding
- Trained a self-supervised autoencoder in PyTorch to compress 512-d embeddings to 64-d latent space

MakeIO 3rd Place (500 participants) | Python, YOLOv8, TensorFlow, PyTorch, Raspberry Pi

Mar 2026

- Built a multimodal edge AI pipeline combining YOLOv8 detection with an audio CNN for real-time wildlife health
- Achieved 0% accuracy degradation vs. supercomputer server baseline at 14.9x lower energy consumption

PUBLICATIONS

Cross-Modal Corroboration for Annotation-Free Wildlife Monitoring

Jun 2026

First Author – CV4Animals Workshop, CVPR 2026

- Developed a self-validating framework finding depth estimation of animals from ground based camera sensors using BioCLIP 2 vision and acoustic classification

SmartWilds: Multimodal Wildlife Monitoring Dataset

Sep 2025

Co-Author – Imageomics Workshop, NeurIPS 2025

- Contributed to a peer-reviewed dataset integrating drone imagery, camera traps, and bioacoustics for conservation-scale computer vision, species detection, and habitat monitoring research